

Proof Architecture in the OpenAI Lean Library: Representation, Locality, and Semantic Transport

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Abstract

Formal proofs often become most informative where they change representation: a graph is encoded by a small matrix, or a random experiment becomes a program reading specified tape bits. The mathematical question is what must be preserved for the next proof step to remain valid. This source-driven review examines that question in six developments from the OpenAI Lean collection: common-base counting, graph switch chains, the symmetric Mahler inequality, the complete Crouzeix bound, extended trace cones, and marked character-variety constructions. We trace the correspondences between representations and identify their downstream use. Across the cases, the required information includes exact coordinates, output laws, quantitative estimates, and local differential data. The comparison explains why an output mean cannot replace a probability law, why ordered form identities and boundary locality have different roles, and why equal subspaces need not preserve a named-coordinate equation. Elementary worked examples isolate these distinctions; a composed success-budget argument connects sampling loss and tape uniformity to an exact finite success count. Comparison with established formalization and verified-programming literature locates the contribution in the concrete composition of these interfaces. The review offers mathematicians and formalizers a route from a theorem statement to the preservation arguments that make its construction work.

Keywords: formalized mathematics; Lean; proof architecture; semantic preservation; probabilistic algorithms; mathematical interfaces.

1 Introduction

A formal proof can cross several mathematical languages before reaching its conclusion. A graph becomes a Boolean matrix, a differential form acquires a continuously bundled representation, or an estimator becomes a program reading a particular sequence of random bits. Each crossing raises the same question: what information must survive for the next argument to remain valid?

We study the proved correspondences that answer this question. We call them preservation interfaces: they specify both the relation between representations and the information available to a later construction. An accuracy event requires the output probability law; differentiation requires suitable local information about a field; an equation in named coordinates requires the coordinate maps themselves. Equivalence of underlying objects is useful only when it supplies the relation that the next step consumes.

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The OpenAI Lean collection offers six detailed cases for this comparison [32]. Common-base counting and graph switch chains show how finite representations support probabilistic guarantees. The Mahler and Crouzeix arguments show how algebraic identities, analytic estimates, and auxiliary domains meet. Trace ideals and marked character-variety diagrams show how an invariant or a change of coordinates can retain enough data to reconstruct and transport a specific object. The first four selected statements are public endpoints; the final two are support results with their own explicit conclusions.

Three questions guide each case. What does the intermediate representation retain? Which theorem connects it to the intended object? Where does that theorem discharge the next step’s hypotheses? The review is intended for mathematicians reading formal developments and for formalizers choosing interfaces. Its contribution is a source-anchored comparison, supported by small examples that explain why a weaker correspondence can fail.

This perspective builds on established accounts of formalization. Lean combines dependent type theory, an extensible elaborator, and a programming language [16]; Mathlib provides a shared mathematical vocabulary [35]. Avigad et al. [2] follow a linear-algebra example through formal proof, generalization, and connection with an existing Mathlib theorem. Commelin and Topaz [12] study abstraction boundaries for mathematical construction and collaboration, using usable specifications as evidence of alignment with informal mathematics. Our comparison follows already-written preservation arguments across several settings.

Large formalizations provide further precedents. The odd-order theorem and Kepler conjecture combined extensive libraries, proof decompositions, and verified computation [19, 20]. Perfectoid spaces and schemes in Lean required carefully chosen objects and interfaces [7, 8]. Against that background, the interest here lies in the particular mathematical relation proved at each transition and the operation it enables. The elementary examples serve that explanatory purpose. Preservation of data, laws, estimates, and local information provides a common reading framework for the cases.

Section 2 describes selection and evidence. Section 3 begins with the overview in Table 1, then develops the six cases. Section 4 isolates their preservation requirements in worked examples. Section 5 brings the comparisons together and considers their use in new formalizations.

2 Sources, selection, and reading method

The unit of comparison is a chain: an initial representation, a preservation statement, and its downstream use. Selection was purposive. A case had to expose all three parts and a mathematically meaningful change of representation. We favored diversity of mathematical setting and preservation requirement over coverage by file count.

For each case we read backwards from the selected conclusion to its definitions and immediate hypotheses, then forwards through the constructions that discharge those hypotheses. We ask which device is inherited from earlier work, which concrete relation is proved in this development, and which later obligation it resolves. The accompanying evidence ledgers retain declaration names, line ranges, source excerpts, and Git blob identifiers; downloaded files were checked against those identifiers.

The discovery process was exploratory. The retained findings record endpoint inspection followed by deeper tracing of candidate interfaces. For example, the irrationality-exponent-of- π and two-point-correlation endpoints were examined but left outside the review because their mechanisms had not been traced at comparable depth. These exclusions reflect the available analysis. The selection supports a comparison of the six cases, not an estimate of how frequent their techniques are throughout the collection.

A central reading distinction is between a conditional interface and its instantiation. An analytic theorem may assume a flux identity or resolvent estimate; the next question is whether the endpoint constructs that input for its stated objects. Similarly, a witness chosen classically has a proved construction, while an extra provider hypothesis remains an input to the theorem. We trace these dependencies directly. The six case studies use static declaration and dependency analysis. The original examples in Section 4 provide a separate executable check of particular arithmetic and finite-law interfaces, using Lean 4.34.1. Repository provenance and checking infrastructure are documented in the data and code availability statement.

Blueprints connect informal arguments to declarations and dependencies [25]; LeanArchitect studies automation of that connection [38]. Our evidence ledgers make the selected interfaces inspectable. The exposition supplies the mathematical equations that explain why their hypotheses are useful.

3 Six mathematical interfaces

Table 1 summarizes the cases in three pairs: probability and computation, analysis, and reconstruction of algebraic data. Each account follows the objects through a preservation statement to its mathematical use. The graph-table calculation and Crouzeix energy cancellation are developed within their cases; Section 4 returns to the other interfaces through smaller examples.

Table 1: Overview of the selected formal statements and the information preserved by each bridge.

Development	Representation bridge	Mathematical work exposed
Common bases Endpoint	Matroid transversals; finite random trees; oracle expressions; physical random tapes.	Output-law preservation joins bit-cost bounds and a fixed-program specification.
Graph switches Endpoint	Fixed-degree graph cells; 2×2 Boolean matrices; conditional averaging operators.	A finite count isolates one exceptional direction per fiber; compensated estimates control its global contribution.
Mahler Endpoint	Ordered shuffle forms; continuously bundled fields; boundary-local extensions.	Exact slot identifications preserve derivatives, signs, and multiplicities; locality makes Stokes applicable.
Crouzeix Endpoint	Amplified polynomial evaluation; rectangular compressions; Fourier coefficient energies.	Finite dual witnesses and positive blocks give a scalar cancellation; outer domains remove strict-containment assumptions.
Trace ideals Support result	Closed ideals; idempotent extended traces; positive cuts and scalar limits.	Algebraic recovery of ideals combines with canonical convergence to identify transported ideals.
All seams Support result	A shared splitting witness; named graded maps; rebased matrices and produced diagram values.	Eight local seam families cover the constructed diagram; equality transport retains its named maps.

3.1 Common bases: from a matroid representation to physical random tapes

The common-bases development connects an approximate counting argument to a fixed oracle machine, including its bit operations and the tape positions from which it obtains randomness [28]. Its organizing choice is to give the mathematical estimator, the finite random experiment, and the compiled program separate semantics, then prove the correspondences between them.

Encoding the counting problem. The mathematical problem is to count the common bases of two equal-rank matroids M_1, M_2 on an enumerated ground set $E = \text{Fin } n$. The first representation replaces two simultaneous base conditions by one base condition. On $E \sqcup E$, set

$$N = M_1 \oplus M_2^*, \quad T(A) = \text{inl}(A) \cup \text{inr}(E \setminus A).$$

The source proves that $T(A)$ is a base of N exactly when A is a base of both input matroids.¹ Equal ranks give $r(N) = n$, while every transversal $T(A)$ has size n . Consequently the deficiency $d(A) = n - r_N(T(A))$ vanishes precisely at the objects being counted. Explicit equivalences between subsets, finite subsets, and transversals transfer both cardinalities and partition sums. If Z is the common-base count, the resulting annealing partition function satisfies

$$Z \leq \sum_{A \subseteq E} q^{d(A)} \leq Z + 2^n q, \quad 0 \leq q \leq 1.$$

This representation identifies the quantity approximated by the sampling argument. It also supplies a useful state space for exchanges: a plain state selects one element from every doubled pair, while a defect state has one empty pair, one full pair, and singleton choices elsewhere. The walk can pass through these paired defects; a hitting construction returns to plain states. The formalization can therefore distinguish the space supporting local moves from the transversals whose weighted count is ultimately estimated.

The next correspondence replaces rank access by independence queries. A greedy oracle routine computes the rank of an enumerated subset; its correctness identifies the oracle-defined deficiency and annealing weights with their mathematical counterparts. The estimator also uses a sampled zero-deficiency witness as a final guard. This gives the exact zero-count behavior later required of every machine execution.

From experiments to a bounded program. Finite random experiments are represented by trees whose internal nodes choose uniformly from a finite index set. A bounded implementation supplies each draw with a finite bank of rejection trials. Its fit condition requires at most M successive draws and $H \geq \lceil \log_2 a \rceil$ bits per trial at every draw of arity a . With D trials per draw, the development then bounds the loss in a $[0, 1]$ -valued observable by $M2^{-D}$. Both depth and width matter: insufficient trial width can force immediate failure regardless of D . A separate relation permits permutations of the outcome indices at each node and proves that these reindexings preserve expectations. Thus the abstract exchange walk and its code can use different finite enumerations while retaining the same output law.

The code layer makes randomness explicit input. Its typed expressions take an input and a random bank, returning an optional value and the unused bank. A realization theorem relates execution on every bank to the corresponding random tree. For a Metropolis step, the construction follows the actual proposal, acceptance calculation, Bernoulli experiment, and lazy branch. The complete estimator composes these experiments with the witness guard.

¹Base correspondence and the rank of the paired matroid.

Compilation supplies two distinct conclusions: the same encoded output and a bound on its cost. The constructed nine-register stack machine implements the expressions, with a structural simulation proof including list folds. Source evaluation returns a value and a charge; successful evaluation gives the same machine output within that charge. A separate recursive certificate bounds intermediate encoding widths, from which polynomials bound steps, oracle calls, and bit operations. Thus output semantics and size control meet at the fixed program, while remaining separately reusable for its preamble and bank-processing postamble.²

From output laws to successful tapes. The final probabilistic interface connects the program to its accuracy guarantee. Let V be the canonical bounded experiment, and let v be the rational output function implemented by the postamble. The certificate asserts

$$\mathbb{E}_w \varphi(v(w)) = \mathbb{E}_b \varphi(V(b)) \quad \text{for every } \varphi : \mathbb{Q} \rightarrow \mathbb{R},$$

where both expectations are finite uniform averages. It also records source execution and nonnegativity, including zero output when $Z = 0$.³ The universal observable interface transports the output distribution through changes of enumeration and bank layout.

The machine correspondence reaches actual tape addresses. For m total gathered bank bits, there is a start offset s , fixed before quantifying over tapes, such that every tape $\beta : \mathbb{N} \rightarrow \{0, 1\}$ produces

$$v(\text{physicalWord}(s, m, \beta)), \quad \text{physicalWord}(s, m, \beta)(i) = \beta(s + 3(m - 1 - i) + 1), \quad i < m.$$

The reversal comes from pushing bits onto a stack; the stride three reflects the sampling loop's instructions. The proof supplies $s + 3m + 1 \leq B$. Injectivity of these positions shows that a uniform length- B tape induces a uniform length- m word.⁴ Here m counts bank bits, whereas B bounds complete machine execution.

The endpoint chooses one program and constants C, a before the input matroids, oracle, and rational parameters $0 < \varepsilon, \delta < 1$. For every exact independence oracle and matroids of full ground set and common supplied rank r , it uses

$$B = C(1 + n + \ell + \lceil \varepsilon^{-1} \rceil + \text{clog}_2 \lceil \delta^{-1} \rceil)^a,$$

where ℓ is the encoded input length and clog_2 is the ceiling binary logarithm. Every infinite tape halts by B , with both resource counters at most B . For a length- B word, $\text{out}(w)$ denotes this output using its zero extension as the tape. Specializing the observable to the accuracy indicator gives

$$\#\{w \in \{0, 1\}^B : (1 - \varepsilon)Z \leq \text{out}(w) \leq (1 + \varepsilon)Z\} \geq (1 - \delta)2^B.$$

The specification packages this success count with the universal resource and zero-output guarantees.⁵ Verified compilation and probabilistic semantic preservation are established methods [5, 6, 23, 24]. Expectation-functional semantics for randomized programs is also established [1]. In Lean, SampCert separates sampling procedures from normalized distribution semantics and proves exact sampler laws [15]. The distinctive evidence here is the concrete integration: the proof carries the counting quantity through oracle code, width bounds, and physical tape coordinates to an exact finite success count. Sections 4.1–4.3 isolate the uniformity, law, and error-budget arguments behind this last step.

²Cost-bearing compiler correctness; polynomial majorant.

³The output certificate.

⁴Physical tape positions and their uniform law; the machine correspondence.

⁵The fixed-program and finite-tape specification.

3.2 Graph switches: a local table and a fiberwise error direction

The common-base case needs preservation of an entire output law. The graph case instead needs control of the interaction between conditional averages. Its switch-chain development connects a four-vertex graph representation to a spectral estimate [33]. At each step the chain stays put with probability $1/2$; otherwise it chooses uniformly a four-vertex set and an ordered pair of distinct perfect matchings on it. It replaces the first matching by the second when the removed edges are present and the added edges absent, and stays put otherwise. Each of the $6\binom{n}{4}$ proposals thus has probability $1/(12\binom{n}{4})$. Its endpoint assumes $n \geq 4$ and a graphical degree sequence d with $d(v) \leq n - 1$. It bounds the mixing time at total-variation threshold $1/4$ by $2n^8$ and, when there is more than one realization, gives spectral gap at least

$$\frac{1}{24n^2\binom{n}{4}}.$$

The disjoint-pair branch of its proof is organized around a small exceptional direction in conditional averaging.

Reducing a graph cell to a matrix count. States are simple graphs on $\text{Fin } n$ realizing d . Fix four distinct vertices, grouped into pairs $a = \{i, j\}$ and $b = \{k, l\}$. A coarse cell fixes the outer graph, the internal edge of each pair, and every outside vertex's total incidence into each pair. The remaining data consist of a 2×2 Boolean matrix M of cross edges and two sets U, W recording the singleton boundary choices at i and k . The source proves that these coordinates determine the graph.⁶

The important feasibility step reduces all four degree equations to preservation of the cross-edge total and two margin equations:

$$\text{row}_0(M) + |U| = c_L, \quad \text{col}_0(M) + |W| = c_R,$$

where c_L, c_R are fixed by the original graph. Explicit forward and inverse constructions identify the fixed-degree graph cell with left and right boundary data together with a compatible matrix. This equivalence preserves the multiplicities needed for uniform conditional averages. In particular, projecting a uniform graph cell onto its boundary data produces weights equal to the number of compatible cross matrices. The matrix count records the entire dependence between the left and right data that remains after the outer graph and incidence totals have been fixed.

There are only sixteen Boolean matrices. Let $c_s(r, t)$ count those with total s , first-row sum r , and first-column sum t . For $s = 2$, finite decision proves the table

$$(c_2(r, t))_{r, t=0}^2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad c_2(r, t) = 2 - p(r) - p(t),$$

where $p(r) = \mathbf{1}_{r \neq 1}$.⁷ For every other total, the table factors into independent margin weights. For total two, the centered covariance factors through one flag on each side: conditional averaging can depend on the left data through the single Boolean flag $p(r)$. This is the dependence that the subsequent analysis tracks.

A short calculation illustrates the exceptional correlation. Among the six uniformly chosen total-two matrices, each off-center margin flag has mean $1/3$, while their product is always zero.

⁶The coarse graph cell; degree feasibility in coordinates.

⁷The finite total-two count.

Their covariance is therefore $-1/9$. This calculation concerns the uniform six-matrix experiment; graph cells also carry boundary multiplicities. It explains why a blanket claim of independent margins or nonnegative local correlation would fail and why the proof isolates a compensated error direction.

Paying for the exceptional correlation. In the finite coordinate model, composing the two conditional averages therefore gives $A + Bp(r)$, with $B = 0$ unless $s = 2$. The graph-cell equivalence transports this formula back to actual graph states. This transport includes a proof that corresponding fibers have equal cardinalities and corresponding sums, and an identification of the flag: $p(r)$ says that i has equal adjacency bits to k and l , meaning both cross edges are present or both are absent.⁸

Work in the real function space on the finite graph state space Ω , with counting inner product $\langle f, g \rangle = \sum_{G \in \Omega} f(G)g(G)$. Fiber averages themselves are divided by the fiber cardinality.⁹ Write E_a for averaging over a pair-resampling fiber. The transported formula expresses the relevant conditional image as a fiber-constant base plus a fiber-constant weight times the equality flag; the weight is supported where the cross total is two. For this purpose a fiber is active when k and l each have exactly one neighbor in a . Subtracting E_a leaves a multiple of the centered flag on each active fiber. These functions form an error subspace whose orthogonal projection is denoted R_{ab} . The description is at most one dimensional on each fiber; the complete graph-space subspace can have many independent fiber components.

This representation supplies the hypotheses of an abstract projection interaction inequality. Conditional averaging is a symmetric idempotent operator, and the error projection fixes the correlated fluctuation just identified. The analytic argument decomposes the state-space function using ranges and orthogonal complements of these projections. This turns the graph-specific conditional formula into precisely the range conditions required by the operator theorem. With G_{ab} the appropriate switch-gain projection, its graph instance yields

$$2\|G_{ab}f\|^2 - \|R_{ab}f\|^2 - \|R_{ba}f\|^2 \leq 2\langle (I - E_a)f, (I - E_b)f \rangle.$$

A separately proved projection budget bounds the sum of squared error norms by the sum of squared gain norms. Summing the compensated inequality over ordered disjoint two-element vertex sets a, b then gives

$$\sum_{a, b: a \cap b = \emptyset} \langle (I - E_a)f, (I - E_b)f \rangle \geq 0.$$

Thus the proof controls the aggregate interaction by accounting explicitly for the exceptional flag directions.¹⁰

The nonnegative disjoint sum combines with triple-generator and local comparison estimates to produce a generator-square bound, then the stated spectral gap and mixing estimate. Computation on finite objects and transport through equivalent representations are established proof methods [11, 21]. The case-specific mathematical choice is to isolate the total-two table's equality flag, identify its centered error space on graph fibers, and compensate those errors in a global operator estimate.

⁸The affine conditional image; its graph interpretation.

⁹See `OrthogonalPairResampling.lean`, lines 191–193, and lines 231–232.

¹⁰The concrete interaction inequality; assembly with the projection budget.

3.3 Mahler: ordered forms and boundary locality

The analytic cases shift the preservation question from finite laws to identities and estimates used in differentiation and integration. The symmetric Mahler development connects exterior algebra, complex analysis, and convex geometry through several representations of the same differential form. Its selected endpoint, `OAI.SymmetricMahler.symmetric_mahler`, states

$$\frac{4^n}{n!} \leq |K| |K^\circ|$$

for $n \geq 1$ and a compact, convex, origin-symmetric set $K \subseteq \mathbb{R}^n$ with nonempty interior. Here volume is ordinary Lebesgue volume and the coordinate polar is literally $K^\circ = \{p : \sum_i p_i v_i \leq 1 \text{ for every } v \in K\}$. The accompanying mathematical argument uses holomorphic maps associated with finite intersections of strips [31]. For the architectural review, the instructive point is how a local transgression identity becomes a numerical flux value usable by the convex-body reduction.

The analytic inputs and the transgression identity. The generic analytic results take a structure named `MassHypotheses`. Its fields require an open domain containing zero, holomorphic components f_j , zero as their unique common zero in the domain, a homogeneous degree- m leading polynomial map G without nonzero common zeros, an actual Euclidean Taylor remainder $f - G = O(\|z\|^{m+1})$, and compact closures of sublevels with $0 < R < 1$ contained in the domain. The structure records analytical inputs. The desired mass inequality and numerical flux value are separate conclusions. For the actual strip maps, `euclideanSpecial_massHypotheses` constructs every field from positive dimension, an injective real measurement matrix, and $m \geq 2$. The final body theorem supplies the resulting homogeneous flux value to a conditional reduction lemma. Its retained inputs are precisely the body hypotheses stated above.¹¹

For the transgression calculation, write the complex dimension as $N = k + 2$, let $q(z) = z$, and set $\tau(H)(z) = \sum_j |H_j(z)|^2$. The source convention is $d^c u(z)[v] = -Du(z)[iv]/4$. On punctured space the degree- m homogeneous map G defines

$$\beta = d^c \log \tau(G) - m d^c \log \tau(q), \quad \alpha_t = m d^c \log \tau(q) + t\beta.$$

The deformation uses the ordered primitive

$$\eta_t = \alpha_t \wedge \beta \wedge (d\alpha_t)^k.$$

The ambient differentiation theorem preserves both terms:

$$d\eta_t = \beta \wedge (d\alpha_t)^{k+1} - \alpha_t \wedge d\beta \wedge (d\alpha_t)^k.$$

It follows from actual Fréchet differentiation, exterior product rules, and $d^2\alpha_t = 0$. Restriction to sphere tangents then kills the first term. The deformation residual has $k + 1$ possible placements of $d\beta$; their verified multiplicity gives

$$\text{residual}_t = -(k + 1)d\eta_t \quad \text{on sphere tangent tuples.}$$

Keeping the ambient identity separate from its tangential consequence makes the vanished term, sign, and multiplicity individually accountable.

¹¹The endpoint and concrete hypothesis construction are `Mahler/MainTheorem.lean`, lines 12–22, and `Mahler/EuclideanSpecialMap.lean`, lines 67–97.

The disappearance of the first term has a concrete linear-algebra proof. At $z \neq 0$, the sphere tangent space has real dimension $2N - 1$ and contains the nonzero circle tangent iz . Homogeneity gives $\beta_z(iz) = 0$ and $(\alpha_t)_z(iz) = m/2$. Rotation invariance, together with this constant circle evaluation, gives $d\alpha_t(iz, v) = 0$. Thus β and $d\alpha_t$ are horizontal in this direction: inserting iz in any slot gives zero.¹² Their top-degree wedge therefore vanishes. The lemma `horizontal_top_eq_zero` proves the general assertion explicitly. An independent top-degree tuple determines a basis. Choose a nonzero basis coordinate of the horizontal vector, replace the corresponding tuple entry by that vector, and expand it in the basis. Alternation kills every term except the selected coordinate times the original evaluation; cancelling the nonzero coordinate proves that evaluation zero. A dependent tuple already vanishes by alternation. The sphere theorem instantiates this argument with the verified tangent dimension and iz .

Identifying two representations of the form. The representation change occurs because the development constructs η_t in two useful ways. One bundles a literal shuffle form through finite basis expansion, supporting regularity proofs by evaluation. The other builds a continuous product from a two-form and continuous wedge powers, supporting differentiation rules. Recursive slot types retain each factor's ordered pair of arguments. Two equivalences flatten these slots to $\text{Fin}(2(k + 1))$ along different routes. A small extensional proof identifies the equivalences; injectivity of the projection from continuous alternating maps to alternating maps then identifies the entire fields. This field equality transports the proved transgression into the analytic theorem expecting the other representation. Thus the intermediate theorem `homogeneousFlux_endpoints_of_transgression` has an explicit local-formula premise, which `homogeneous_flux_endpoints` discharges.¹³

Each representation carries different proof obligations. The literal shuffle description keeps permutation signs and factor order available to algebraic rewrites. The continuously bundled field has a normed-space structure, so its regularity and derivative can be expressed through the existing calculus API. Finite basis expansion connects these obligations: smoothness of the finitely many evaluated coefficients supplies smoothness of the bundled form. The source obtains every finite regularity order on punctured space from the analytic hypotheses, including the C^2 order used below. Identifying the fields also identifies their derivatives as functions valued in continuous alternating maps. This is stronger than checking their values on a chosen tangent tuple, which suffices only at the later integration boundary. The proof therefore uses two distinct congruence principles at appropriate stages: whole-field equality for differentiation and tangential agreement for oriented density. Keeping those principles separate clarifies which information each mathematical operation consumes.

Applying Stokes near the boundary. A second interface controls locality. The primitive is C^2 on punctured space, while the flux argument needs regularity near the sphere. `exists_close_d_set_form_extension` takes a form ω that is C^2 on an open set U containing a closed set K . Smooth separation gives a function χ equal to one near K and zero near U^c . The field $\tilde{\omega} = \chi\omega$ is globally C^2 : use the product rule inside U and its eventual equality to zero outside U . Near K it is eventually equal to ω , so their Fréchet derivatives, and hence exterior derivatives, agree there. For K the sphere, coordinate Stokes applies to $d\tilde{\omega}$ and $d^2\tilde{\omega} = 0$ gives zero boundary flux. Tangent-density congruence then transfers that conclusion to the original residual.¹⁴

¹²The normalization and circle evaluations are in the d^c definition, lines 12–13, and the deformation and circle identities, lines 14–101.

¹³See the homogeneous endpoint, lines 12–45. The underlying tangent identity is proved in the tangent transgression theorem, lines 65–97.

¹⁴The extension and exact-form flux theorem are `Mahler/ClosedBoundaryStokes.lean`, lines 18–63.

Zero residual flux makes the deformation's path flux constant by the fundamental theorem of calculus. Comparing endpoints supplies $(\pi m)^N$; the small-sphere limit and special strip-map reduction use that value in the body inequality. Dimension one is handled by a separate circle calculation. The mathematical devices are familiar. Mathlib already provides continuous alternating maps, exterior differentiation, and $d^2 = 0$ under smoothness assumptions [36]. Kudryashov's earlier Lean formalization already transfers a divergence theorem between integration theories and derives Cauchy–Goursat under carefully chosen differentiability assumptions [22]. Its rectangular domains give a direct comparison for integration interfaces. The distinctive step here identifies ordered exterior algebra with continuously bundled fields and combines that equality with a locality theorem that imposes regularity where the integral uses it. These interfaces expose and discharge the substantive premises without requiring regularity at the singular center. The elementary boundary-form example in Section 4.4 separates the ambient and tangential information used at these two stages.

3.4 Crouzeix: compression, energy, and strict neighborhoods

The finite-matrix Crouzeix endpoint states an amplified norm estimate with constant two. For an n -by- n complex matrix A , m -by- m coefficients B_k , and $n, m > 0$, `OAI.DirectCrouzeix.complete_crouzeix` states

$$\left\| \sum_{k=0}^d A^k \otimes B_k \right\| \leq 2 \sup_{z \in W(A)} \left\| \sum_{k=0}^d z^k B_k \right\|.$$

Write $F(z) = \sum_{k=0}^d z^k B_k$; $F(A)$ denotes the amplified evaluation on the left. The source uses the operator 2-norm, literal matrix Kronecker products, and $W(A) = \{\langle u, Au \rangle : \|u\| = 1\}$. Its endpoint quantifies over arbitrary A and coefficient matrices. The accompanying argument [30] is organized around boundary analysis; the formal implementation makes particularly clear how positivity and Fourier energy combine to produce the constant two.

Closing the compressed energy estimate. The local interface `analytic_domain_core` accepts a polynomial sequence b_k with $b_0 = 1$, a continuous boundary parametrization, and a matrix resolvent field $R(t)$. Its hypotheses give finite Faber expansions, lower and upper coefficient-energy bounds, exact coefficients $\hat{R}(-k) = b_k(A)$, vanishing positive Fourier coefficients, and $R + R^* \geq 0$. These are conditional inputs to the local theorem. `exterior_collar_bound` constructs them from exterior conformal data and supporting halfplanes before applying the core. In particular, the positive resolvent and coefficient bounds are outputs at this geometric stage, rather than retained assumptions of the public endpoint.

For the normalized core, assume $\|F(h(t))\| \leq 1$ on the boundary, where h is its parametrization. Let T be the amplified polynomial evaluation and $\gamma = \|T\| > 0$; the zero case is separate. A unit singular pair satisfies $Tx = \gamma y$ and $T^*y = \gamma x$. The supporting lemma obtains a norm attainer by compactness, then proves the adjoint relation through a squared-distance calculation. Reshaping vectors indexed by $\text{Fin}(n) \times \text{Fin}(m)$ produces rectangular matrices X, Y . All boundary integrals below use normalized circle measure. The proof forms

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY, \quad a = \int \|M(t)\|_{\text{HS}}^2 dt.$$

A cross-boundary pairing estimate proves $a > 0$. The two singular relations produce the diagonal bounds

$$\int \|P + P^*\|_{\text{HS}}^2 \leq \frac{4a}{\gamma^2}, \quad \int \|Q + Q^*\|_{\text{HS}}^2 \leq \frac{4a}{\gamma^2}.$$

The normalization $b_0 = 1$ identifies the zero Fourier coefficients:

$$\hat{P}(0) = X^*X, \quad \hat{Q}(0) = Y^*Y, \quad \hat{L}(0)^* = \hat{M}(0).$$

The first two are Hermitian; the last is the zero-frequency premise of the cross-energy comparison. All four compressions inherit the vanishing of the positive Fourier coefficients of R . Fourier identities then give $a \leq \int \|M + L^*\|_{\text{HS}}^2$. Compressing $R + R^*$ by the concatenated columns $[X \ Y]$ yields a positive block matrix with these diagonal and off-diagonal blocks. For a positive block matrix $D = \begin{pmatrix} U & Z^* \\ Z & V \end{pmatrix}$ and $J = \text{diag}(I, -I)$, positivity of D and JDJ gives

$$0 \leq \text{Re tr}(DJDJ) = \|U\|_{\text{HS}}^2 + \|V\|_{\text{HS}}^2 - 2\|Z\|_{\text{HS}}^2.$$

Integrating this estimate gives

$$2 \int \|M + L^*\|_{\text{HS}}^2 \leq \int \|P + P^*\|_{\text{HS}}^2 + \int \|Q + Q^*\|_{\text{HS}}^2.$$

Thus $2a \leq 8a/\gamma^2$, and cancellation of $a > 0$ gives $\gamma \leq 2$.¹⁵

Where the sharp energy weights come from. Two details explain why this reduction works sharply. First, `singular_diagonal_estimates` tests both ordered products FG and GF , where G is a test polynomial. The forward and adjoint singular relations turn them into different diagonal compressions, preserving the noncommutative order of matrix coefficients. The dual step is explicit: from bounds on every finite test family C_k , `weighted_coefficient_dual` substitutes $C_k = w_k^{-1} \overline{P_k}$ in the project's bilinear entry pairing. Both pairing and test energy become $s = \sum_k \|P_k\|_{\text{HS}}^2 / w_k$, leaving $s^2 \leq bs$ and hence $s \leq b$. Nonnegative bounded partial sums subsequently give the infinite sequence estimate. This keeps the optimizing witness finite and concrete.

Second, the weights are derived by boundary analysis. A Faber series splits into a nonnegative Fourier series and a remainder with no nonnegative coefficients. Orthogonality gives its energy as the sum of coefficient energy and remainder energy. The remainder is also the action of a nonnegative kernel with both marginals one on the nonconstant series. A weighted squared-norm inequality and Fubini prove kernel contraction. Consequently $w_0 = 1$ and $w_k = 2$ for $k > 0$. The resulting half-weighted coefficient tail fits the exact Fourier identity

$$\int \|P + P^*\|_{\text{HS}}^2 = 4 \left(\|\hat{P}(0)\|_{\text{HS}}^2 + \frac{1}{2} \sum_{k \geq 1} \|\hat{P}(-k)\|_{\text{HS}}^2 \right).$$

Its factor four at frequency zero uses Hermiticity of $\hat{P}(0)$; the negative coefficients contribute in paired positive and negative frequencies.¹⁶

A further interface supplies matrix-valued Faber expansions from scalar polynomials. A scalar sequence with its k th polynomial of degree k is mapped into polynomials over the matrix coefficient ring by sending scalars to scalar matrices. Its leading coefficients are units. The implementation then applies Mathlib's degree-filtered spanning theorem for polynomial sequences and extracts a finite matrix-coefficient witness from membership in that span. This gives the exact finite expansion consumed by the core. Evaluation and Kronecker-evaluation identities carry the witness to the two concrete matrix calculations. The established polynomial basis theorem is reused across the change of coefficient ring, while explicit transport lemmas preserve the order of multiplication. Together with rectangular reshaping, this explains how the scalar boundary analysis supplies estimates uniformly in the matrix coefficient size, the defining feature of the complete bound.

¹⁵The four compressions and closing cancellation appear in `DirectCrouzeix/DomainCore.lean`, lines 15–117.

¹⁶See `BoundaryEnergy.lean`, lines 57–88, and `PositiveBlocks.lean`, lines 22–62.

Removing the auxiliary domain assumptions. The final interface removes strict-domain geometry. For every compact, nonempty convex $K \subseteq \mathbb{C}$, `exists_exponential_outer` constructs a compact sublevel of a finite sum of exponentials of affine support inequalities. A finite direction net controls approximation; four coordinate directions ensure boundedness. The sublevel contains K strictly and lies in any prescribed neighborhood of K . In the endpoint, set $M = \sup_{W(A)} \|F\|$ and $\rho = M + \varepsilon/2$. Compactness places such an outer sublevel inside the open set $\{z : \|F(z)\| < \rho\}$. The scaled domain estimate gives $\|F(A)\| \leq 2M + \varepsilon$, and removing $\varepsilon > 0$ gives the desired bound. The final assumptions are positive matrix sizes and arbitrary coefficients; smooth strict containment is supplied by an auxiliary construction.¹⁷

Singular vectors, dual witnesses, positive blocks, and approximation are established tools. Their precise composition is the architectural lesson: amplification becomes rectangular compression, the two polynomial orders become two energy bounds, and positivity closes a scalar cancellation. As a comparison, the Crouzeix–Palencia proof centers the Cauchy transform of conjugate holomorphic functions and obtains the complete constant $1 + \sqrt{2}$ [13]. The comparison identifies different proof interfaces, without asserting priority for the individual devices used here.

3.5 Recovering ideals from extended traces and canonical limits

The final pair of cases concerns the mathematical data retained by an invariant or a coordinate change. The trace-cone development reconstructs actual closed ideals from an analytic invariant and then proves that an equivalence of invariants transports two distinguished ideals. The mathematical setting is the cone of extended, lower-semicontinuous traces, studied as an invariant of C^* -algebras by Elliott et al. [17]. The selected support results concern this ideal correspondence. Their hypotheses are nonunital complex C^* -algebras with compatible order and star structures; they impose no density condition on the finite domain of a trace [34].

Recovering the ideal order algebraically. An `ExtendedTraceA` is a function on positive elements with values in $[0, \infty]$, together with proofs of vanishing at zero, additivity, positive homogeneity, traciality, and lower semicontinuity. For a closed two-sided star ideal I , the code constructs the extended trace

$$\iota_I(a) = \begin{cases} 0, & a \in I, \\ \infty, & a \notin I, \end{cases} \quad a \geq 0. \quad (1)$$

To establish additivity for positive a, b , heredity of closed ideals gives $a + b \in I$ exactly when both summands belong to I , proving additivity. The equivalence $x^*x \in I \iff x \in I$ and star invariance give traciality; closed sublevel sets give lower semicontinuity. The existence theorem in `FiniteIdeal.lean` therefore supplies a trace satisfying the full interface, from which the later definition chooses a witness.

The reverse construction starts with the finite-square domain of a trace:

$$\mathcal{N}_\tau = \{x \in A : \tau(x^*x) < \infty\}, \quad I_\tau = \overline{\mathcal{N}_\tau}^{\|\cdot\|}. \quad (2)$$

The source proves that $\mathcal{N}_\tau$ is a nonunital star subalgebra absorbing multiplication on both sides, then takes its topological closure. Thus I_τ denotes a concrete closed ideal. If $\tau + \tau = \tau$, every value is either zero or infinity. Finite-trace positive elements approximate every positive element of I_τ ; lower semicontinuity extends their zero values to that limit, yielding $\tau = \iota_{I_\tau}$. Conversely, $I_{\iota_I} = I$.

¹⁷See `DirectCrouzeix/OuterApproximation.lean`, lines 59–122, and `CompleteBound.lean`, lines 65–97.

These inverse laws establish `closedIdealEquivIdempotent`, an equivalence between closed ideals and additive-idempotent traces.

The ideal order is also encoded by addition:

$$I \subseteq J \iff \iota_I + \iota_J = \iota_I. \quad (3)$$

Consequently, any additive bijection e of the two extended-trace cones already induces an order isomorphism Φ of their closed ideals, with $\iota_{\Phi(I)} = e(\iota_I)$. Neither continuity nor compatibility with scalars is required for this part. The subsequent analytic argument identifies which ideals this order isomorphism assigns to a given trace. This separation makes the strength of each interface field visible.

Identifying the transported ideals by limits. The analytic step uses a cut-regularized notion of convergence. Write $(a - \varepsilon)_+$ for the positive cut defined by continuous functional calculus. The development proves

$$a \in I_\tau \iff \forall \varepsilon > 0, \quad \tau((a - \varepsilon)_+) < \infty, \quad a \geq 0. \quad (4)$$

Its canonical convergence predicate for a filtered family (τ_i) and a candidate limit σ requires, for every positive a and every $\varepsilon > 0$,

$$\limsup_i \tau_i((a - \varepsilon)_+) \leq \sigma(a) \leq \liminf_i \tau_i(a). \quad (5)$$

The two inequalities have different roles: the left regularizes the argument by a positive cut; the right uses the original element. Their combination, together with recovery of a lower-semicontinuous trace from its cuts, gives uniqueness of a canonical limit along a proper filter.¹⁸

In this sense the source proves

$$r\tau \longrightarrow \iota_{I_\tau} \quad (r \downarrow 0), \quad r\tau \longrightarrow \iota_{Z_\tau} \quad (r \rightarrow \infty), \quad (6)$$

where Z_τ is a closed ideal whose positive elements are exactly those on which τ vanishes. The first limit depends on the closure and cut characterization in (4). Whenever $a \in I_\tau$ but $\tau(a) = \infty$, every positive scalar multiple still has value $(r\tau)(a) = \infty$, although $\iota_{I_\tau}(a) = 0$. A claimed pointwise scalar-zero limit would therefore miss the recovered ideal. Finite values on all positive cuts reconcile this behavior with (5). The scalar-infinity limit, by contrast, is first proved pointwise using the trace that is zero on τ 's zero set and infinity elsewhere.

The ideal weights, cut-regularized topology, and these two scalar limits are classical parts of the trace-cone analysis of Elliott et al. [17, Sections 3.1–3.2]. The source makes their formal dependencies explicit: additive structure recovers the ideal order, while convergence and scalar compatibility identify the two ideals attached to an arbitrary trace. The review's comparison concerns that separation and its implementation. Section 4.5 gives a sequence-space example in which the canonical scalar-zero limit and the pointwise limit differ.

Finally, a `TraceConeEquiv` packages an additive bijection, preservation of zero and positive scaling, and canonical convergence in both directions. For each scalar limit, the endpoint proof maps the family through e , rewrites using scalar compatibility, and invokes uniqueness of canonical limits. Injectivity of $I \mapsto \iota_I$ then yields

$$\Phi(I_\tau) = I_{e(\tau)}, \quad \Phi(Z_\tau) = Z_{e(\tau)}.$$

¹⁸The convergence predicate is defined in `TraceCone/Basic.lean`, lines 65–69; uniqueness is proved in `IdealTransport.lean`, lines 177–190.

The displayed endpoint proofs use forward convergence preservation; the inverse field belongs to the larger equivalence interface. The illuminating architecture is the chain from constructed ideal weights, through an algebraic reconstruction, to analytic recognition by limits. It makes the ideal correspondence an actual transported object, rather than an unspecified provider assumption. This is a specific use of the abstraction boundaries familiar from large formalizations [7, 12].

3.6 Named graded maps across all seams of a produced diagram

The character-variety case proves that values produced by a marked surface construction satisfy every seam equation. The selected theorem, `producedMarkedValues_allSeams`, starts with an existing field-valued solution of the original diagram. Its hypotheses include properness of the diagram, a maximal-rank condition on the facet at the chosen seam, and positive genus of the parent facet. The conclusion refers to the specific diagram and values constructed from that solution [26]. The key issue is preserving named graded data while changing coordinates and subdividing the surface.

The diagram consists of surface pieces, called facets, joined along seams; frames specify the graded identifications at those joins. At the selected seam, the construction inserts a refined band determined by a simultaneous splitting of the two input frames. It retains the old boundary circles, appends a mirror boundary at the parent facet, and introduces a genus-zero disk for each fresh strip; the parent genus drops by one. The all-seams theorem checks compatibility of the matrix values on this resulting combinatorial diagram. This roadmap explains the inherited, mirror, and branch families used below.¹⁹

The information in a named graded map. The seam equation is stronger than equality of flags. For a graded index set with grade function a and a frame e , let

$$F_k^e = e\{u : u_i = 0 \text{ whenever } a(i) \geq k\}, \quad q_k^e(v) = (e^{-1}v)|_{\{i:a(i)=k\}}.$$

Here q_k^e retains the named coordinates of grade k . The predicate `SameFramedFlagae` requires

$$F_k^e = F_k^f, \quad q_k^e(v) = q_k^f(v) \quad \text{for all } v \in F_{k+1}^e, \quad (7)$$

for every k . Thus the comparison retains the maps representing the successive graded pieces, rather than only their dimensions or underlying subspaces.²⁰ In the produced seam equations, the two frames are obtained from evaluations of the seam's left and right matrix words. Losing these named maps during a change of basis would weaken the theorem's meaning. The one-dimensional frame example in Section 4.6 illustrates this distinction without the surface construction.

Keeping one witness across the construction. The intermediate construction records precisely the data needed to prevent that loss. An `IdentifiedBand` contains a common frame for a simultaneous splitting, a rank shape, and row and column identifications of its graded fibers. Its prefix and named-coordinate laws relate those identifications to the two input frames. The source first proves that such a record exists using the named band grades and rank equalities. It then chooses a witness. The opaque declaration `cutGeneratorWitness` has this construction as its body; opacity hides reduction, rather than adding an unproved existence assumption. Classical choice here supplies coherent mathematical data, without claiming an effective procedure for computing the frame.

¹⁹See the constructor description in `Cutting/MarkedDiagram.lean`, lines 79–104.

²⁰See the definition of `SameFramedFlag`.

Crucially, the same chosen record supplies the shape, row ranks, column ranks, and marked frames of the produced diagram. Those fields also determine its punctures and values. Independently choosing compatible dimensions would not provide these shared identifications. This illustrates the difference between packaging a structure and carrying a sufficiently strong specification through it. Records and algebraic interfaces are standard techniques [3, 18]; the case-specific payload is the simultaneous preservation of the named graded maps in (7).

From local seams to the produced values. The global seam proof is assembled from eight local families: the chosen old seam, the other inherited seams, the first and internal original seams, the last and internal mirror seams, and the positive and negative branch seams. Exhaustiveness is proved using a surjective equivalence describing the grafted diagram, followed by sum, endpoint, and Boolean cases. Consequently, the universal quantifier over new seams follows from a structural coverage theorem. In the inherited family, coordinate equivalences identify the new frames with the old named seam frame; grade-preservation lemmas make these equivalences suitable for the full predicate. The original solution then supplies the seam equation. The local proof transports coordinate maps, not merely a statement that the old and new ranks agree.

There remains a second transport at the endpoint. The general `markedCut_allSeams` theorem proves the eight-family result for the assembled marked-cut data. The produced values use matrices rebased along the equality of the selected seam rank and its parent-facet rank. `rebaseUnit` performs this equality-induced change of matrix size. The endpoint proves that the chosen and rebased generator matrices agree, uses the mirror-rebasing theorem, and transports the resulting equality through the side-value constructor and then `valuesFromPorts`. Finally, `framedSeamEquations_of_values_eq` transfers the general seam predicate to the exact produced values. These equalities tie the compatibility proof to the constructed representation.

The boundary assumptions also distinguish two nearby statements. The all-seams endpoint has boundary indices b_0, i_0 , but requires no proof that this occurrence is the chosen seam. The later `producedMarkedSolution`, which assembles surface, seam, and vertex equations into a complete solution, additionally assumes

$$D.\text{boundarySide}\langle D.\text{bandCutParent}(q), b_0, i_0 \rangle = \langle q, \text{none} \rangle.$$

That extra hypothesis is used for the surface component. Keeping these endpoints distinct exposes the exact scope of the selected support theorem. The architectural lesson is the combination of a common witness, exhaustive local families, and equality transport that preserves the named mathematical content through the final construction.

4 Worked examples: why the stronger interface matters

The first three examples unpack the common-base case: selecting random bits, preserving the output law, and composing an error budget. The remaining examples return to boundary locality, trace limits, and named coordinates. Each isolates the information needed by a later operation without the surrounding formal development.

4.1 Injective tape positions preserve a uniform experiment

Proposition 1. *Let $p : \{0, \dots, m-1\} \rightarrow \{0, \dots, B-1\}$ be injective. If β is uniform on $\{0, 1\}^B$, then $(\beta(p(0)), \dots, \beta(p(m-1)))$ is uniform on $\{0, 1\}^m$. Every word has exactly 2^{B-m} preimages.*

Proof. Prescribing the m selected bits leaves precisely $B - m$ coordinates free. Each assignment to these remaining coordinates determines one preimage. Thus every word has the same number of preimages, and its probability is $2^{B-m}/2^B = 2^{-m}$. $\square$

For the common-base machine, the relevant map is $p(i) = s + 3(m - 1 - i) + 1$. Reversal and multiplication by three preserve distinctness. For $i < m$, its value is less than B whenever $s + 3m + 1 \leq B$. The latter bound is stronger than needed for the selected positions because it also accounts for the surrounding execution. The review’s Lean file proves these arithmetic facts for arbitrary natural s, m, B , including the empty-word case.

The index map is defined by

```
def physicalIndex (start m : Nat) (i : Fin m) : Nat :=
  start + 3 * (m - 1 - i.val) + 1
```

The proof uses the bounds stored in `Finm` to justify subtraction arithmetic. This matters: natural-number subtraction is truncated, so injectivity would fail on unrestricted natural indices. A range type is part of the mathematical interface, not a cosmetic annotation.

For comparison, if $p(0) = p(1)$, the words 01 and 10 have no preimages. Reading fair individual bits does not by itself produce an independent word. Proposition 1 explains why the source proves injectivity of the physical positions before transporting the success probability. The Python companion exhaustively checks 15 small choices of (s, m, B) against the exact fiber count.

4.2 Observables determine a finite law; its mean does not

Proposition 2. *Let μ, ν be probability measures on a finite set X . Then $\mathbb{E}_\mu \varphi = \mathbb{E}_\nu \varphi$ for every $\varphi : X \rightarrow \mathbb{R}$ if and only if $\mu = \nu$.*

Proof. For each $x \in X$, take $\varphi = \mathbf{1}_{\{x\}}$ to obtain $\mu(\{x\}) = \nu(\{x\})$. Summing singleton masses gives equality on every subset. The converse follows from the definition of finite expectation. $\square$

In a uniform finite experiment, these expectations are normalized sums with integer multiplicities. The Lean companion proves the corresponding count statement: equality of unnormalized sums against every natural-valued observable determines each multiplicity. If two sampling spaces have different cardinalities $N, N' > 0$, equality of their output laws instead means $N'c(x) = Nc'(x)$ for each output x . The companion also proves this cross-multiplied statement from equality of normalized observables. For example, multiplicities $(1, 1, 1)$ and $(2, 2, 2)$ describe the same law, although their unnormalized counts differ. It then checks the measures on $X = \{0, 1, 2\}$ with multiplicities

$$(1, 1, 1) \quad \text{and} \quad (0, 3, 0).$$

Both have total mass three and mean one after normalization. Their probabilities of $\{0\}$ are $1/3$ and zero. Thus preservation of the expected numerical output would be insufficient for an accuracy event. The universal observable certificate in Section 3.1 supports exactly the stronger conclusion required: substitute the indicator of the desired relative-error interval.

This interpretation also explains why the source allows reindexing of draw outcomes. A permutation can change the result returned for a particular numbered bank without changing

its distribution under uniform sampling. Law equality is the appropriate interface between those experiments; the deterministic compiler layer separately preserves the execution result for a supplied bank.

4.3 From a bounded experiment to a finite success count

The preceding two interfaces can be composed with a quantitative error bound. This gives a concrete way to read the final accuracy statement of Section 3.1.

Proposition 3. *Let X be finite, adjoin a failure symbol $\perp$, and let U be an ideal X -valued output. Let V be a bounded implementation driven by a uniform m -bit word, taking values in $X \cup \{\perp\}$. For $\varphi : X \rightarrow [0, 1]$, set $\varphi(\perp) = 0$. Suppose*

$$\mathbb{E}\varphi(V) \geq \mathbb{E}\varphi(U) - \eta \quad \text{for every such } \varphi.$$

Let $\eta, \delta_0 \geq 0$ and $A \subseteq X$ satisfy $\Pr(U \in A) \geq 1 - \delta_0$. If a machine obtains its word from fixed injective positions in a uniform B -bit tape and returns V on that word, then

$$\#\{\text{successful } B\text{-bit tapes}\} \geq (1 - \delta_0 - \eta)2^B,$$

where success means returning an element of A .

Proof. Choose $\varphi = \mathbf{1}_A$ to obtain $\Pr(V \in A) \geq 1 - \delta_0 - \eta$. Proposition 1 shows that the tape projection has the required uniform word law. Multiplying its success probability by 2^B gives the count. $\square$

For a single accuracy event, only its indicator is needed; the all-observables hypothesis makes the estimate reusable for other events. In the source’s bounded draw-tree implementation, the fit condition requires at most M successive draws and $H \geq \lceil \log_2 a \rceil$ trial bits for every draw arity a . With D trials per draw, $\eta = M2^{-D}$, with failed runs assigned observable value zero.²¹ For a desired failure budget $0 < \delta < 1$, choosing $\delta_0 \leq \delta/2$ and $M2^{-D} \leq \delta/2$ suffices. When $M > 0$, one may take $D \geq \max\{0, \lceil \log_2(2M/\delta) \rceil\}$. The required tape length and execution bound must still be proved separately. Thus the accuracy budget, the uniformity of the gathered bits, and the deterministic machine specification are distinct inputs to a single final guarantee.

A tight finite example joins the layers. Let the ideal probabilities of a, b be $6/8, 2/8$, and assign the eight implementation words to five copies of a , two of b , and one failure. The observable loss is exactly $\varphi(a)/8$, hence at most $\eta = 1/8$. For $A = \{a\}$, $\delta_0 = 1/4$ and the lower bound is $5/8$. Embedding the three-bit word at positions 7, 4, 1 in a ten-bit tape gives exactly $5 \cdot 2^7 = 640$ successful tapes out of 1024. The Python companion enumerates those tapes and all event indicators. This fixture illustrates the composed bound independently of the upstream rejection sampler.

4.4 Boundary values and ambient derivatives

The Mahler case uses local equality to identify ambient derivatives. On $\mathbb{R}^2$, the one-forms $\omega = 0$ and $\eta = (x^2 + y^2 - 1) dx$ agree at every point of the unit circle. Their ambient exterior derivatives need not agree: $d\eta = -2y dx \wedge dy$, which is nonzero at $(0, 1)$. Equality on a neighborhood does imply equality of the ambient derivatives. The Mahler cutoff argument uses this sufficient condition to transfer its ambient derivative; agreement on tangent arguments suffices for the later comparison of integration densities. Intrinsic differentiation after pullback to the circle uses different data: both pulled-back forms, and their exterior derivatives, vanish. Mathlib’s differential-form interface explicitly uses the derivative of a form-valued field [36].

²¹See `CommonBases.lean`, `run_lower`.

4.5 A cut-sensitive limit that is not pointwise

Consider the commutative C^* -algebra c_0 of complex sequences tending to zero, with pointwise operations and the supremum norm. On its positive cone define

$$\tau(a) = \sum_{j=1}^{\infty} a_j \in [0, \infty].$$

The trace is additive, positively homogeneous, and tracial. It is lower semicontinuous because it is the supremum of its continuous finite partial sums. Finite-support sequences belong to $\mathcal{N}_\tau$, so the closed finite ideal I_τ in (2) is all of c_0 .

For every positive $a \in c_0$ and $\varepsilon > 0$, the sequence $(a - \varepsilon)_+$ has finite support. Consequently

$$\lim_{r \downarrow 0} r\tau((a - \varepsilon)_+) = 0, \quad 0 \leq \liminf_{r \downarrow 0} r\tau(a).$$

These are precisely the two inequalities in (5) with limit trace $\iota_{c_0} = 0$. But for $a_j = 1/j$ one has $\tau(a) = \infty$, hence $r\tau(a) = \infty$ for every $r > 0$. The pointwise limit at this element is infinity. The finite ideal and canonical limit are nevertheless unambiguous.

This classical example makes the reason for the positive cuts concrete [17]. Closure admits elements with infinite trace, while every fixed positive cut is controlled by a finite value. Replacing the source’s convergence predicate by pointwise convergence would therefore alter the theorem.

4.6 Equal flags and different named coordinates

The character-variety case preserves more than the flag of subspaces. Two frames of a one-dimensional vector space can induce the same flag but different named coordinate maps. If $e(u) = uv$ and $f(u) = 2uv$ over a field of characteristic zero, both give the flag $0 \subset \text{span}(v)$, while $e^{-1}(v) = 1$ and $f^{-1}(v) = 1/2$. The all-seams predicate must therefore retain the named graded maps when those maps occur in the seam equation. The coordinate equation consumes information that equality of the underlying flag has forgotten. This is the small-scale reason for retaining named graded maps throughout the surface construction.

5 Discussion: preservation and reuse

5.1 What the comparison reveals

Across the six cases, the downstream operation determines the strength of the required correspondence. An accuracy probability depends on the output law. A named-coordinate equation depends on the coordinate map. Ambient differentiation uses local information about a field. An optimized inequality depends on the normalization and weights of its estimates. The four recurring requirements—preservation of exact data, laws, estimates, and local differential information—can occur together within one development.

The most revealing comparisons concern a change in what the proof needs. The common-base construction preserves a distribution through reindexing, then preserves execution for a fixed input bank. Mahler first identifies whole form-valued fields so that it can differentiate them, then uses agreement on tangent tuples to compare boundary integrals. The trace-cone construction recovers ideal order from addition alone, then uses convergence and scalar compatibility to recognize the ideals of a particular trace. In each case, following the next operation explains why the interface has the fields it does.

Two cases also show how a small calculation can govern a large estimate. For graph switches, the total-two matrix table identifies an exceptional direction whose aggregate contribution is paid for by a projection budget. For Crouzeix, rectangular compressions and two polynomial orders produce energy bounds that close at the constant two. The important relation is between the local calculation and the global quantity it controls.

These patterns have substantial antecedents. Verified compilers establish semantic preservation between source and implementation [23, 24]; probabilistic program logics reason about distributions and observations [5, 6]. Algebraic hierarchy design and specification-driven collaboration treat interfaces as mathematical objects [3, 12, 18]. The contribution of this comparison lies in showing the specific mathematical payload of those interfaces and how their composition supports the selected proofs.

5.2 Using the comparison in a new formalization

Start with the intended downstream statement. Write the equation, inequality, or event probability that the next step requires. Then identify what the proposed representation changes: multiplicities, coordinate order, multiplication order, or a neighborhood of definition. The preservation statement should connect those changed data to the quantity the next theorem consumes.

The examples in Section 4 make this choice concrete. For an accuracy event, preserving a mean is insufficient; its indicator expectation is enough, and a full output-law identity makes the interface reusable for other events. For ambient differentiation, neighborhood agreement is sufficient, while the later boundary integral may require only agreement on tangent tuples. A named seam equation needs its coordinate map. Proposition 3 then shows how several such obligations can meet: sampling loss, tape uniformity, and deterministic execution give one finite success count.

Finally, follow the construction of the hypotheses for the actual inputs. The Mahler strip maps supply the analytic data, while the Crouzeix outer domains supply strict containment and the required estimates. In the marked-surface construction, using one splitting witness for the diagram and its values keeps their identifications coherent. These examples suggest reusable interfaces for a library: a small local model with a measure-preserving correspondence, a sufficient semantic relation, and explicit construction of the inputs that relation requires.

5.3 Scope and further work

Selection favored explicit interfaces that could be traced in depth. It supports the present comparisons but does not estimate their frequency or development cost across the collection. The source analysis establishes what the inspected declarations state and how their constructions connect. A full assessment of the headline mathematical results additionally requires rebuilding their dependencies and examining the underlying arguments, with the endpoint and support-result distinctions retained.

For a mathematician, these cases offer a route into substantial formal source: locate the next operation, identify what it consumes, and follow the theorem that preserves that information. For a formalizer, they point to interfaces worth extracting into smaller libraries. The worked examples provide a modest version of that extraction, exposing a mathematical dependency in a form that can be inspected, reused, and tested.

Data and code availability

The accompanying source package contains the LaTeX manuscript, verified reference records, declaration evidence ledgers, original Lean examples, and Python checks. The Lean examples use Lean 4.34.1 and import only `Lean`; their compiler log records the axioms reported for the five named theorems. The Python checks enumerate tape fibers and Boolean matrix counts using exact integers. The package README gives reproduction commands and identifies the pinned external sources. Public source links in the manuscript resolve to the reviewed repository revision.

The source inventory is the complete Git tree of `openai/math`, inspected on 8 October 2026.²² It contains 122,875 files with the `.lean` extension, including supporting modules. The separate formalization catalogue lists 173 source records and 200 main-result bindings [27]. These counts describe different units; none is a count of the developments studied in depth here.

Proof checking supplies complementary evidence. The collection provides Comparator challenge configurations [29]; the separately inspected checker implements statement comparison, dependency and axiom inspection, kernel replay, and an optional external checker [9].²³ The distinction between constructing a proof and checking it has a long history [4, 14]; Lean typechecker verification provides a contemporary example [10]. Library maintenance addresses consistency at another level [37]. Understanding why a formal statement captures the intended mathematics still requires reading its objects and hypotheses.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

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²²Reviewed revision: `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Public source links refer to that revision.

²³Checker source revision: `ca04cfc72b550331658ec314bf47685281bfd4bf`. This pin records our infrastructure reading; the collection’s installation instructions do not specify that revision.

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